System Model and Problem Formulation

Pilot-Based Channel Estimation for an Uplink Rydberg Atomic MIMO Receiver

1 System Description

We consider the uplink of a single-cell multi-user system in which K single-antenna users transmit to a Rydberg atomic receiver. The receiver comprises N receive elements arranged as a uniform linear array (ULA). Channel state information is acquired from P pilot symbols transmitted by each user over consecutive time instants.

The indices used throughout are summarised below.

Index	Range	Meaning
n	$1, \ldots, N$	receive element
k	$1, \ldots, K$	user
p	$1, \ldots, P$	pilot instant
ℓ	$1, \ldots, L_k$	propagation path of user k

Terminology. We refer to the N units as *receive elements*, each consisting of one atomic ensemble (a vapour cell) interrogated by a single photodetector. We deliberately avoid the term "N Rydberg atoms": each element contains $N_a \approx 5 \times 10^5$ participating atoms, the measured quantity is an ensemble-averaged optical response, and the quantum projection noise variance depends explicitly on N_a. The symbols N and N_a therefore denote distinct quantities.

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2 Modelling Assumptions

The receiver architecture adopted here employs a reference radio-frequency (RF) field at the same carrier frequency as the user signals, together with an electromagnetically induced transparency (EIT) Autler–Townes (AT) splitting readout. The assumptions underlying the model are listed below, with their provenance indicated.

#	Assumption	Provenance
A1	K single-antenna uplink users, one receiver	[1]–[3]
A2	ULA with element spacing $d = \lambda/2$	geometry from [2]; spacing fixed here
A3	Far-field, plane-wave, specular multipath with L_k resolvable paths per user	[2]

#	Assumption	Provenance
A4	Narrowband operation: $B_s(N-1)d/c_0 \ll 1$, so path delays across the array act as pure carrier phase	introduced here
A5	Co-polarised array: the projection $\mu_{eg} \cdot \epsilon = \wp$ is a known real constant, identical for all (n, k, ℓ)	introduced here
A6	Each element samples the RF field at a point (spatial extent $\ll \lambda$)	introduced here
A7	Negligible mutual coupling and negligible RF re-radiation between elements	introduced here
A8	Two-level atomic model, on resonance ($\delta = 0$), rotating-wave approximation	[1], [2]
A9	Reference field at the same carrier ω_c, line-of-sight dominant, with known amplitude and phase at every element	[1], [2]
A10	Block fading: **H** remains constant over the P pilot instants	introduced here
A11	Perfect synchronisation; the pilot matrix **S** is known at the receiver	[2]
A12	Noise enters as $\mathcal{CN}(0, \sigma^2)$ inside the magnitude operation	[1], [2]
A13	Path gains $\alpha_{\ell,k} \sim \mathcal{CN}(0, \beta_k/L_k)$, independent across paths	introduced here
A14	Complex-valued pilots and complex reference gain	introduced here
A15	The atomic conversion gain c is known	introduced here

Remark on assumption A5. In [2] the polarisation projection is indexed by the element, and is drawn independently per element in the accompanying simulations. This is incompatible with the separable array response that underpins the low-rank structure exploited in the same work. We therefore fix the projection to be constant across elements, which restores the array manifold structure.

Remark on assumption A8. The two-level resonant model is adopted as a communication-theoretic abstraction of a same-carrier-reference receiver whose readout is the EIT–AT splitting interval. It is not equivalent to the four-level susceptibility model of [3], in which the two-level abstraction is criticised as an incomplete description of a realistic Rydberg receiver. The abstraction is valid in the regime where the RF field is near-resonant with a single Rydberg transition, the readout is amplitude-only, and the atomic response is monotone in the RF Rabi frequency. Extension to the four-level model is beyond the present scope.

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3 Array Geometry

The receive elements are located on the x-axis at positions $\mathbf{p}_n = (n-1)d\,\hat{\mathbf{x}}$, $n = 1, \dots, N$, with

$$d = \frac{\lambda}{2}, \qquad \lambda = \frac{c_0}{f_c},$$

where f_c denotes the carrier frequency and c_0 the speed of light.

The ℓ-th path of user k arrives from angle $\theta_{\ell,k} \in [-\pi/2, \pi/2]$, measured from array broadside. For $\theta_{\ell,k} > 0$, element n is farther from the source than element 1 by $(n-1)d\sin\theta_{\ell,k}$. We define the associated spatial frequency

$$\psi_{\ell,k} \triangleq \frac{2\pi d}{\lambda}\sin\theta_{\ell,k} \overset{d=\lambda/2}{=} \pi\sin\theta_{\ell,k} \in [-\pi, \pi].$$

Since $d \le \lambda/2$, the mapping $\theta \mapsto \psi$ is injective on $[-\pi/2, \pi/2]$, so that no spatial aliasing occurs and angles of arrival remain uniquely identifiable.

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4 Propagation Channel

Under A4, the differential path delays across the array affect only the carrier phase. The baseband channel coefficient between user k and element n is therefore

$$\boxed{h_{n,k} = \sum_{\ell=1}^{L_k} \alpha_{\ell,k}\, e^{-\jmath(n-1)\psi_{\ell,k}}}$$

where the terms are defined as follows.

Symbol	Meaning	Model
$\alpha_{\ell,k}$	complex path gain, comprising path loss, scattering, transmit amplitude, and the carrier phase referenced to element 1	$\mathcal{CN}(0, \beta_k/L_k)$

Symbol	Meaning	Model
$\theta_{\ell,k}$	angle of arrival, measured from broadside	$\mathcal{U}[-\pi/2, \pi/2]$
β_k	large-scale receive gain of user k, inclusive of transmit power	design parameter
L_k	number of resolvable paths of user k	user-dependent

The normalisation in A13 yields $\mathbb{E}|h_{n,k}|^2 = \beta_k$ for every n, and consequently $\mathbb{E}\|\mathbf{h}_k\|_2^2 = N\beta_k$, so that the array gain scales linearly in N.

Defining the array manifold vector

$$\mathbf{a}(\theta) = \left[1,\ e^{-\jmath\psi(\theta)},\ \dots,\ e^{-\jmath(N-1)\psi(\theta)}\right]^{\mathsf{T}} \in \mathbb{C}^N, \qquad \|\mathbf{a}(\theta)\|_2^2 = N,$$

the channel vector of user k admits the compact representation

$$\mathbf{h}_k = \sum_{\ell=1}^{L_k} \alpha_{\ell,k}\, \mathbf{a}(\theta_{\ell,k}) = \mathbf{A}_k \alpha_k \in \mathbb{C}^N, \qquad \mathbf{A}_k \in \mathbb{C}^{N\times L_k},\ \ \alpha_k \in \mathbb{C}^{L_k},$$

and the multi-user channel matrix is obtained by stacking the user channel vectors columnwise,

$$\boxed{\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_K] \in \mathbb{C}^{N\times K}.}$$

Interpretation. The entry $h_{n,k}$ is the complex baseband gain from the transmit-antenna terminal of user k to the co-polarised component of the RF electric field at the phase centre of element n, evaluated at carrier f_c, per unit-power transmitted symbol, aggregated over all L_k paths and phase-referenced to element 1. It contains no atomic or optical quantity.

Parameter count. Treated as unstructured, $\mathbf{H}$ possesses $2NK$ real degrees of freedom. Under the geometric model it is described by $3\sum_k L_k$ real parameters, comprising two per complex path gain and one per angle of arrival.

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5 Atomic-Domain Channel

The atomic response is governed by the Rabi frequency rather than by the field amplitude in volts per metre. We therefore introduce the atomic conversion gain

$$c \ \triangleq\ \frac{\wp}{\hbar} \qquad [\text{rad} \cdot \text{s}^{-1} \text{ per V} \cdot \text{m}^{-1}], \qquad \wp = |\mu_{eg} \cdot \hat{\mathbf{e}}|,$$

with μ_{eg} the transition dipole moment and $\hat{\mathbf{e}}$ the field polarisation unit vector. Under A5, c is a known real positive constant, identical across elements, users and paths. The atomic-domain channel is then

$$\boxed{\mathbf{G} \ \triangleq\ c\,\mathbf{H} \ \in\ \mathbb{C}^{N\times K}, \qquad g_{n,k} = \frac{\wp}{\hbar} \sum_{\ell=1}^{L_k} \alpha_{\ell,k}\, e^{-\jmath(n-1)\psi_{\ell,k}}.}$$

We retain the distinction between **H** and **G** deliberately. The matrix **G**, with units of rad $\cdot$ s^{-1} per unit symbol, is the quantity to which the receiver responds, whereas **H** is dimensionless and is determined solely by the propagation environment. The two differ by a known real scalar, so estimation of one is equivalent to estimation of the other and the normalised mean-square error is identical for both. This separation is not made in [1] and [2], where the dipole moment, the factor $1/\hbar$ and the polarisation projection are absorbed into the channel definition.

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6 Pilot Signalling and Reference Field

At pilot instant p, user k transmits the known symbol $s_{k,p}$ with $\mathbb{E}|s_{k,p}|^2 = 1$. We collect the pilots as

$$\mathbf{s}_p = [s_{1,p}, \dots, s_{K,p}]^\mathsf{T} \in \mathbb{C}^K, \qquad \mathbf{S} = [\mathbf{s}_1, \dots, \mathbf{s}_P] \in \mathbb{C}^{K \times P}.$$

Per A14, the pilots are complex-valued, $s_{k,p} \sim \mathcal{CN}(0,1)$. Real-valued pilots would leave the conjugate ambiguity $\mathbf{G} \leftrightarrow \overline{\mathbf{G}}$ unresolved, as discussed in Section 10.

A single reference source transmits the known symbol $s_{b,p}$ at the same carrier frequency ω_c over a line-of-sight path from a fixed and known geometry at angle ϑ. Applying the development of Section 4 with a single path yields

$$\boxed{b_{n,p} = c\,\alpha_b\,s_{b,p}\,e^{-\jmath(n-1)\psi_b}, \qquad \psi_b = \frac{2\pi d}{\lambda}\sin\vartheta,}$$

$$\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_P] \in \mathbb{C}^{N \times P},$$

where **B** is fully known at the receiver, with α_b complex-valued and $b_{n,p} \neq 0$ for all (n,p).

We define the reference-to-signal ratio

$$\mathrm{RSR} \;\triangleq\; \frac{\mathbb{E}|b_{n,p}|^2}{\mathbb{E}|g_{n,k}s_{k,p}|^2},$$

which we treat as a design parameter of the system rather than as a nuisance quantity.

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7 Received RF Field

The RF electric fields generated by the users and by the reference source superpose linearly at each element, this being a consequence of Maxwell's equations in free space rather than an approximation. Collecting all elements and pilot instants,

$$\boxed{\mathbf{E} = \mathbf{GS} + \mathbf{B} + \mathbf{W} \;\in\; \mathbb{C}^{N \times P}, \qquad \mathrm{vec}(\mathbf{W}) \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_{NP}),}$$

or, element by element,

$$E_{n,p} = \sum_{k=1}^{K} g_{n,k} s_{k,p} + b_{n,p} + w_{n,p}.$$

The noise variance is $\sigma^2 = \sigma_E^2 + \sigma_S^2$, where $\sigma_S^2 = 4\pi^2/(T_i T_r N_a)$ denotes the quantum projection noise with T_i the integration time, T_r the Rydberg-state relaxation time and N_a the number of participating atoms per element, and σ_E^2 accounts for environmental contributions. Following A12, **W** represents an equivalent pre-measurement noise aggregating the background RF field and the referred readout noise. We note that placing the readout noise inside the magnitude operation is an approximation inherited from [1] and [2]: the background RF field is physically additive at the field level, whereas photon shot noise and quantum projection noise arise downstream at the optical readout.

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8 Atomic Measurement Model

Under A8, the two-level system driven on resonance and treated within the rotating-wave approximation exhibits an excited-state population $\sin^2(\Omega_{n,p}t/2)$, where the RF Rabi frequency at element n and instant p is

$$\Omega_{n,p} = \left|E_{n,p}\right|.$$

The mapping $E \mapsto |E|$ is invariant under $E \to Ee^{\jmath\phi}$ for any real ϕ, and it is at this stage that the complex phase of the received field is lost. The physical origin of this loss is that the RF carrier phase is not referenced against any oscillator at ω_c within the receiver: the readout occurs at optical frequencies, and the reference field, sharing the carrier frequency of the users, adds coherently inside the magnitude rather than producing a beat note against the signal.

The Rabi frequency is read out through the EIT–AT splitting interval $\Delta f_{n,p} = (\lambda_c/\lambda_p)\Omega_{n,p}/2\pi$, with λ_p and λ_c the probe and coupling wavelengths. Rescaling the detected splitting yields the observation matrix

$$\boxed{\mathbf{Z} = \left|\mathbf{GS} + \mathbf{B} + \mathbf{W}\right| \ \in \ \mathbb{R}_+^{N\times P},}$$

with $|\cdot|$ applied elementwise. The complete signal chain is summarised as

$$\underbrace{h_{n,k}}_{\text{propagation}} \longrightarrow \underbrace{\mathbf{E} = \mathbf{GS} + \mathbf{B} + \mathbf{W}}_{\text{RF field at the array}} \longrightarrow \underbrace{\Omega = |\mathbf{E}|}_{\text{atomic response}} \longrightarrow \underbrace{\Delta f \propto \Omega}_{\text{optical readout}} \longrightarrow \underbrace{\mathbf{Z}}_{\text{estimator input}}$$

in which the phase is present up to the atomic response and is destroyed thereafter, the reference field **B** serving to convert relative phase into observable amplitude.

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9 Problem Formulation

Given the observation matrix $\mathbf{Z}$, the known pilot matrix $\mathbf{S}$ and the known reference matrix $\mathbf{B}$, the channel estimation problem is

$$\boxed{\hat{\mathbf{G}} = \arg\min_{\mathbf{G}\in\mathbb{C}^{N\times K}} \left\|\mathbf{Z} - \left|\mathbf{G}\mathbf{S} + \mathbf{B}\right|\right\|_F^2.}$$

This constitutes a *biased phase retrieval* problem: non-convex, magnitude-only, and endowed with a known additive bias that resolves the phase ambiguity. It is neither linear estimation nor generic nonlinear estimation, and it differs from classical unbiased phase retrieval in that the bias alters both the ambiguity structure and the attainable performance.

Since the noise is complex Gaussian and enters inside the magnitude, each observation is Rician distributed. Writing $\lambda_{n,p} = (\mathbf{G}\mathbf{S} + \mathbf{B})_{n,p}$,

$$p(z_{n,p}; \mathbf{G}) = \frac{2z_{n,p}}{\sigma^2} \exp\left(-\frac{z_{n,p}^2 + |\lambda_{n,p}|^2}{\sigma^2}\right) I_0\left(\frac{2z_{n,p}|\lambda_{n,p}|}{\sigma^2}\right),$$

with $I_0(\cdot)$ the zeroth-order modified Bessel function of the first kind, giving the maximum-likelihood formulation

$$\hat{\mathbf{G}}_{\mathrm{ML}} = \arg\max_{\mathbf{G}} \sum_{n=1}^{N}\sum_{p=1}^{P}\left[-\frac{|\lambda_{n,p}|^2}{\sigma^2} + \log I_0\left(\frac{2z_{n,p}|\lambda_{n,p}|}{\sigma^2}\right)\right].$$

The least-squares criterion is the high signal-to-noise-ratio limit of the maximum-likelihood criterion; the two diverge as the signal-to-noise ratio decreases.

Substituting the geometric representation of Section 4 yields the structured formulation

$$\min_{\{\theta_{\ell,k}\},\{\alpha_{\ell,k}\}} \left\|\mathbf{Z} - \left|c\,[\mathbf{A}_1\alpha_1, \dots, \mathbf{A}_K\alpha_K]\mathbf{S} + \mathbf{B}\right|\right\|_F^2,$$

which is posed over $3\sum_k L_k$ real parameters rather than $2NK$.

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10 Identifiability and Pilot Length

Three ambiguities are inherent to the measurement model, and each is resolved by a corresponding design condition.

Ambiguity	Resolved by
Per-element phase $\mathbf{g}_n \to \mathbf{g}_n e^{\jmath\phi_n}$, comprising N free phases that destroy the array structure	known $\mathbf{B} \neq \mathbf{0}$
Conjugate ambiguity $\mathbf{G} \leftrightarrow \overline{\mathbf{G}}$	complex $\mathbf{S}$ and complex α_b
Angular aliasing	$d \leq \lambda/2$

We emphasise the first of these. In the absence of a reference field, the observation at element n depends only on the n-th row of $\mathbf{G}$, so that each row is determined only up to an arbitrary phase. Since the array manifold structure resides across rows, N free row phases would render the geometry unusable. The reference field is therefore not merely beneficial but necessary.

The observation matrix supplies NP real measurements, whereas **G** contains $2NK$ real unknowns. As the rows decouple in the unstructured formulation, the counting bound applies per row and yields

$$P \geq 2K$$

as a necessary condition, together with **S** of full row rank. The structured formulation additionally requires $N > \max_k L_k$ for the angles to be identifiable.

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11 Limiting Regime

The model admits a tractable limiting regime that we record for completeness, and which we treat as a special case rather than as the model itself. When the reference dominates, $|b_{n,p}| \gg |(\mathbf{GS})_{n,p} + w_{n,p}|$, a first-order expansion gives, with $\mathbf{Y} \triangleq \mathbf{Z} - |\mathbf{B}|$ and $\Psi = e^{-\jmath \angle \mathbf{B}}$,

$$\mathbf{Y} \;\approx\; \Re\{\Psi \circ (\mathbf{GS})\} + \bar{\mathbf{N}}, \qquad \bar{n}_{n,p} \sim \mathcal{N}(0, \sigma^2/2).$$

The mapping $\mathbf{G} \mapsto \Re\{\Psi \circ (\mathbf{GS})\}$ is $\mathbb{R}$-linear though not $\mathbb{C}$-linear. Introducing

$$\breve{\mathbf{g}}_n = \begin{bmatrix} \Re\{\mathbf{g}_n\} \\ \Im\{\mathbf{g}_n\} \end{bmatrix} \in \mathbb{R}^{2K}, \qquad \varphi_{n,p} = \begin{bmatrix} \Re\{\psi_{n,p}\mathbf{s}_p\} \\ -\Im\{\psi_{n,p}\mathbf{s}_p\} \end{bmatrix} \in \mathbb{R}^{2K},$$

the problem separates across elements as $\mathbf{y}_n = \Phi_n \breve{\mathbf{g}}_n + \bar{\mathbf{n}}_n$ with $\Phi_n \in \mathbb{R}^{P \times 2K}$, admitting the closed-form solution

$$\hat{\breve{\mathbf{g}}}_n = \left(\Phi_n^{\mathsf{T}} \Phi_n\right)^{-1} \Phi_n^{\mathsf{T}} \mathbf{y}_n, \qquad P \geq 2K.$$

We note that the linearised problem therefore possesses an exact closed-form solution, and that iterative descent on the same objective can at best attain it. The exact model of Section 8 remains the formulation adopted in this work; the reference-to-signal ratio governs the transition between the two regimes.

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12 Summary of Notation and Dimensions

Symbol	Space	Description
$\mathbf{a}(\theta)$	$\mathbb{C}^{N \times 1}$	array manifold vector
$\mathbf{A}_k$	$\mathbb{C}^{N \times L_k}$	steering matrix of user k
α_k	$\mathbb{C}^{L_k \times 1}$	path gain vector of user k
$\mathbf{h}_k$	$\mathbb{C}^{N \times 1}$	propagation channel of user k

Symbol	Space	Description
$\mathbf{H}$	$\mathbb{C}^{N\times K}$	multi-user propagation channel
c	$\mathbb{R}_{>0}$	atomic conversion gain
$\mathbf{G}$	$\mathbb{C}^{N\times K}$	atomic-domain channel
$\mathbf{s}_p$	$\mathbb{C}^{K\times 1}$	pilot vector at instant p
$\mathbf{S}$	$\mathbb{C}^{K\times P}$	pilot matrix
$\mathbf{b}_p$	$\mathbb{C}^{N\times 1}$	reference field at instant p
$\mathbf{B}$	$\mathbb{C}^{N\times P}$	reference field matrix
$\mathbf{W}$	$\mathbb{C}^{N\times P}$	equivalent pre-measurement noise
$\mathbf{E}$	$\mathbb{C}^{N\times P}$	total RF field
$\mathbf{Z}$	$\mathbb{R}_{+}^{N\times P}$	observation matrix
Ψ	$\mathbb{C}^{N\times P}$	unit-modulus reference phase
$\mathbf{Y}$	$\mathbb{R}^{N\times P}$	debiased observation

The dimensional consistency of the model is verified below.

Expression	Dimensions	Result
$\mathbf{A}_k\alpha_k$	$(N\times L_k)(L_k\times 1)$	$N\times 1$
$\mathbf{G}\mathbf{s}_p$	$(N\times K)(K\times 1)$	$N\times 1$
$\mathbf{GS}$	$(N\times K)(K\times P)$	$N\times P$
$\mathbf{GS}+\mathbf{B}+\mathbf{W}$	three terms of $N\times P$	$N\times P$
$\lvert\mathbf{GS}+\mathbf{B}+\mathbf{W}\rvert$	elementwise	$N\times P$
$\Psi\circ(\mathbf{GS})$	Hadamard product	$N\times P$
$\Phi_n\breve{\mathbf{g}}_n$	$(P\times 2K)(2K\times 1)$	$P\times 1$

References

[1] M. Cui, Q. Zeng and K. Huang, "Towards Atomic MIMO Receivers," *IEEE Journal on Selected Areas in Communications*, vol. 43, no. 3, Mar. 2025.

[2] B. Xu, J. Zhang, Z. Chen, B. Cheng, Z. Liu, Y.-C. Wu and B. Ai, "Channel Estimation for Rydberg Atomic Receivers," *IEEE Wireless Communications Letters*, vol. 14, no. 9, Sep. 2025.

[3] T. Gong, C. Yuen, C. M. S. See, M. Debbah and L. Hanzo, "Rydberg Atomic Quantum Receivers for the Multi-User MIMO Uplink," arXiv:2501.18382.